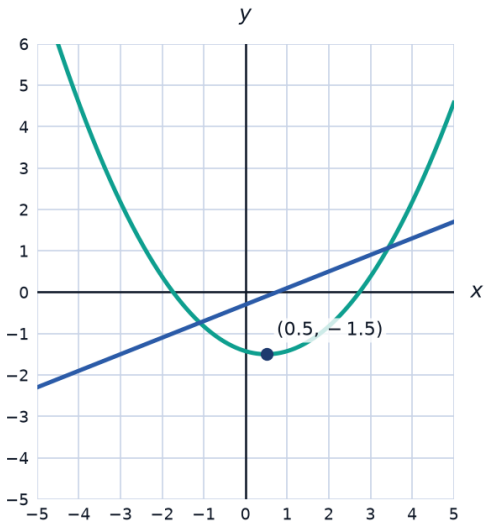
Energy, Work & Power

ClipSAT Printable Practice Suite (clipsat.org) • Topic 14.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A force of 42 N acts at 30° to the direction of motion of a body that moves 27 m. Find the work done, leaving your answer in exact (surd) form.



2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A body of mass 11 kg moving at 7 m/s has work 356 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

[Graphs & Transformations · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[Kinematics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

[Normal Distribution · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Permutations & Combinations · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

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7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Circular Measure · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Statistics & Probability · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In how many ways can 5 people be seated in a row if two particular people (A and B) must not sit next to each other?

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A data set has mean 12 and standard deviation 4. A new value of 20 is added to $n = 9$ existing values. Without finding the new variance, find the new mean.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Rotate the Cartesian axes to eliminate the cross-product term in $5x^2 - 4xy + 8y^2 = 36$. Find the angle of rotation θ, standard form in the rotated system (x', y'), and the lengths of the semi-major and semi-minor axes.

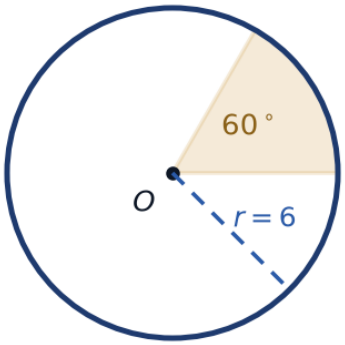
12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Practice 4.2: Bivariate data for study hours x and exam mark y across 10 students yield $\bar{x} = 14.5$, $\bar{y} = 72.0$, and Pearson's correlation coefficient $r = 0.88$. The regression line of y on x has equation $y = 2.4x + c$. (a) Find the value of c. (b) Interpret the slope 2.4 in context. (c) Estimate the exam mark of a student who studies 18 hours.

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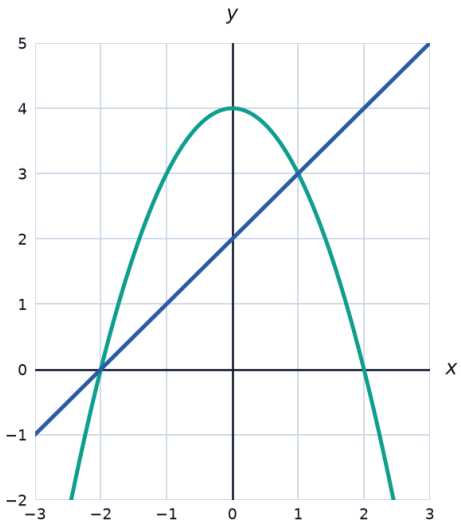
13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve the trigonometric equation $2\sin^2\theta + 3\cos\theta - 3 = 0$ for $0 \le \theta \le 2\pi$, giving exact radian answers.



14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Solve the logarithmic equation $\log_3(x+5) + \log_3(x-1) = 2 + \log_3(2)$ for all real values of x.

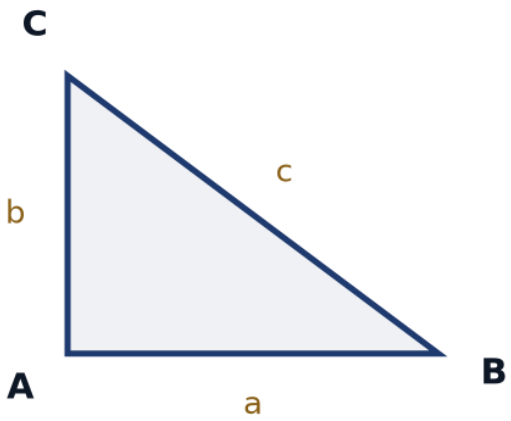


Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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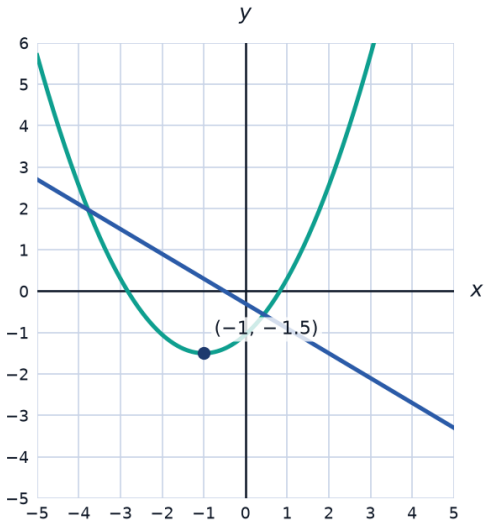
15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 10.01 Series & Binomial Theorem

In ΔABC, $a = 7$, $b = 9$, and $c = 10$. Find the measure of the smallest angle to the nearest tenth of a degree, and compute the area of ΔABC using Heron's formula.



16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 11.01 Kinematics

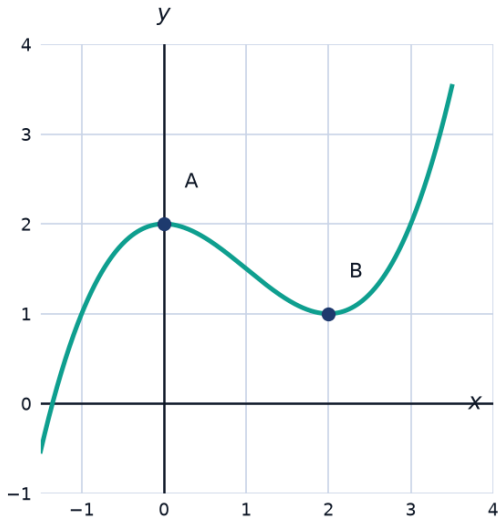
If $\sin\theta = \frac{3}{5}$ with θ terminating in Quadrant I, find the exact values of $\cos\theta$, $\tan\theta$, and $\sin(2\theta)$.



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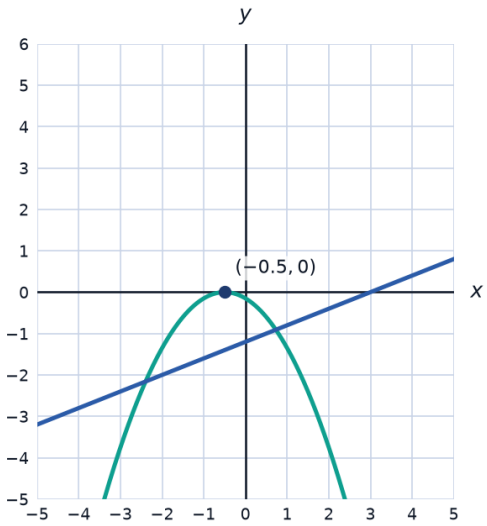
17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 12.01 Forces & Equilibrium

The sum of the first n terms of an arithmetic series is given by $S_n = 4n^2 - 3n$. (a) Find the first term u_1 and the second term u_2. (b) Find the common difference d. (c) Find an explicit expression for the nth term u_n, and calculate u_{25}.



18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 13.01 Newton's Laws

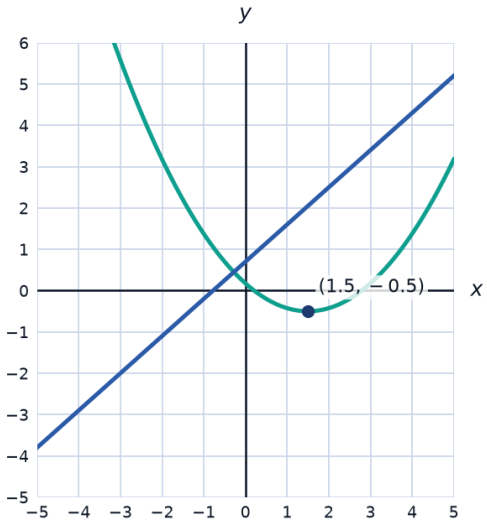
Find the infinite sum of the geometric series $12 + 8 + \frac{16}{3} + \frac{32}{9} + \ldots$. Verify the condition for convergence.



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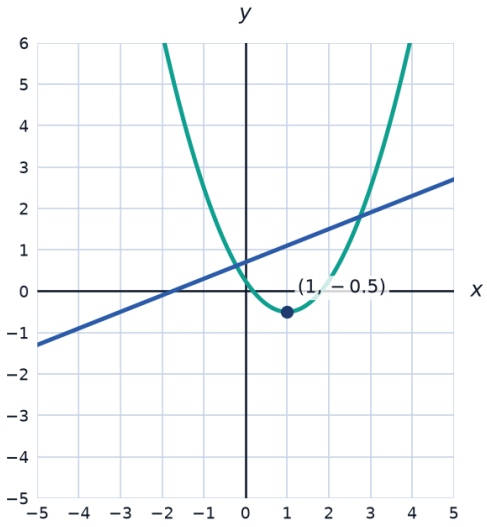
19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 15.01 Representation of Data

Convert the polar equation $r = 4\cos\theta$ to Cartesian rectangular form, classify the curve, and find its center and radius.



20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 16.01 Permutations & Combinations

Convert the polar coordinate point $P\left(6, \frac{4\pi}{3}\right)$ to exact Cartesian coordinates (x, y).



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A force of 60 N pulls a box across a horizontal floor through a distance of 8 m. The force acts at an angle of 60° above the horizontal. Find the work done by the force.

a. 240 J

b. 480 J

c. 415.7 J

d. 120 J

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A crate of mass 15 kg is lifted vertically at a constant speed through a height of 4 m. Taking $g = 10\ \text{m/s}^2$, what is the work done against gravity?

a. 600 J

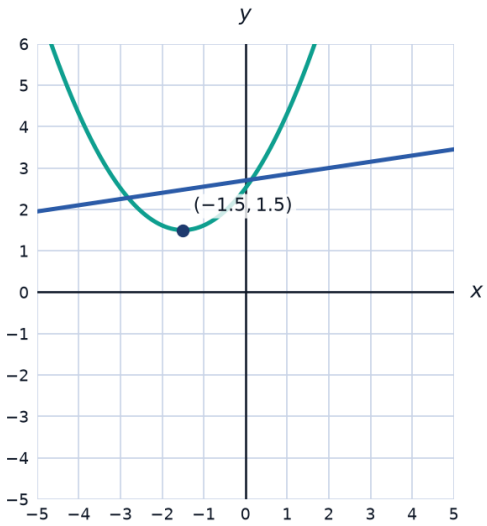
b. 150 J

c. 300 J

d. 60 J

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A box is pushed 5 m along a horizontal floor against a constant frictional resistance of 35 N. What is the work done against friction?



a. 175 J

b. 150 J

c. 350 J

d. 70 J

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Calculate the kinetic energy of an athlete of mass 70 kg running at a speed of 8 m/s.

a. 2240 J

b. 4480 J

c. 560 J

d. 280 J

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A rollercoaster car of mass 500 kg starts from rest at the top of a frictionless hill of height 20 m. Taking $g = 10\ \text{m/s}^2$, find its speed at the bottom of the hill.

a. 20 m/s

b. 10 m/s

c. 40 m/s

d. 14.1 m/s

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A bullet of mass 0.02 kg traveling at 300 m/s penetrates a wooden block to a depth of 0.15 m before stopping. What is the average resistive force exerted by the wood?

a. 6000 N

b. 3000 N

c. 9000 N

d. 12000 N

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A skier of mass 60 kg starts from rest at the top of a slope 30 m high. When reaching the bottom, her speed is 20 m/s. Taking $g = 10\ \text{m/s}^2$, how much energy was lost due to friction?

a. 6000 J

b. 12000 J

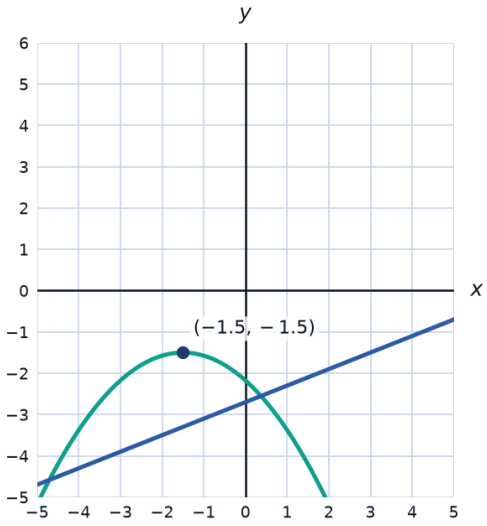
c. 18000 J

d. 3000 J

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A package of mass 4 kg slides down a rough incline of length 10 m and vertical drop 6 m. If the resistive force is a constant 8 N, find the speed of the package at the bottom (starting from rest, $g = 10$ m/s^2).



a. $\sqrt{80} \approx 8.94$ m/s

b. 10 m/s

c. 8 m/s

d. 12 m/s

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A particle is projected up a rough incline (30°) with initial speed 10 m/s. If $\mu = 0.2$ and $g = 10$ m/s^2, how far up the plane does the particle travel before coming to rest?

a. ≈ 7.43 m

b. 10.0 m

c. 5.50 m

d. 8.25 m

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A car of mass 1200 kg travels along a level road at a constant speed of 30 m/s. The engine is working at 45 kW. What is the total resistance to motion?

a. 1500 N

b. 1200 N

c. 1350 N

d. 2000 N

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The same car (mass 1200 kg, engine working at 45 kW) faces the same resistance of 1500 N while traveling at 20 m/s. What is its acceleration at this instant?

a. $0.625\ \text{m/s}^2$

b. $0.75\ \text{m/s}^2$

c. $1.00\ \text{m/s}^2$

d. $0.50\ \text{m/s}^2$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A truck of mass 5000 kg ascends an incline of 1 in 20 ($\sin\alpha = 1/20$) at a steady speed of 15 m/s. The engine power is 60 kW. Taking $g = 10\ \text{m/s}^2$, find the resistance to motion.

a. 1500 N

b. 2500 N

c. 4000 N

d. 2000 N

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A force of 8 N acts on a body, moving it a distance of 7 m in the direction of the force. Find the work done.

a. 56

b. 63

c. 112

d. 49

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A car of mass 1000 kg increases speed from 10 m/s to 20 m/s. The work done is:

a. 50 kJ

b. 100 kJ

c. 150 kJ

d. 200 kJ

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

An engine develops a power of 12483 W while a vehicle moves at a constant 19 m/s. Find the driving force exerted.

a. 657

b. 647

c. 1314

d. 667

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the change in gravitational potential energy of a body of mass 18 kg raised through a height of 15 m. Take $g = 9.8$ m/s^2.

a. 5292.00

b. 2664.00

c. 2646.00

d. 2628.00

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the kinetic energy of a body of mass 16 kg moving at 7 m/s.

a. 376

b. 408

c. 392

d. 784

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 13.01 Newton's Laws

In how many ways can 8 different books be arranged in a row on a shelf?

a. 45808

b. 35108

c. 40625

d. 40320

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 15.01 Representation of Data

In how many ways can 6 different people be seated in a row of 6 chairs?

a. 720

b. 817

c. 800

d. 725

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 16.01 Permutations & Combinations

Find the gradient of the normal to $y = 3x^2 + 2x$ at $x = -1$.

a. -3/4

b. -1/4

c. 5/4

d. 1/4

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 17.01 Discrete Random Variables

A discrete random variable X has distribution $P(X = 1) = 1/10, P(X = 2) = 2/5, P(X = 3) = 1/2$. Find $\mathrm{Var}(X)$.

a. 9/13

b. 11/25

c. 26/37

d. 9/10

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 18.01 Binomial Distribution

Given $X \sim N(62, 5^2)$, find the value a such that $P(X < a) = 0.05$, correct to 2 decimal places.

a. 51.28

b. 56.28

c. 70.22

d. 53.78

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 19.01 Normal Distribution

A histogram class has width 5 and frequency 21. Find the frequency density.

a. 16/5

b. 42/5

c. 26/5

d. 21/5

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

Find the equation of the circle with centre $(3, -5)$ and radius 2.

a. $(x-3)^2 + (y+5)^2 = 4$

b. $(x+3)^2 + (y-5)^2 = 4$

c. $(x-3)^2 + (y+5)^2 = 2$

d. $(x-3)^2 + (y+5)^2 = 8$

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Graphs & Transformations

Find the x-coordinates of the stationary points of $y = 3x^3 - 9/2x^2 - 54x$.

a. $x = -2$ or $x = -3$

b. $x = -2$ or $x = 3$

c. $x = -1$ or $x = 3$

d. $x = 2$ or $x = 3$

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